

V2 Expansion Port Quick Reference

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Orientation used here

This reference matches the photo orientation where:

- the ****expansion port is at the top****
- the ****72-pin cartridge connector is at the bottom****

In that exact view:

- reading starts at the ****far left****
- first pair at left is `01 / 48`
- last pair at right is `24 / 25`

Pin order in this photo orientation

```
```text
haut : 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22
 23 24
bas : 48 47 46 45 44 43 42 41 40 39 38 37 36 35 34 33 32 31 30 29 28 27
 26 25
```
```

V2 useful pins only

```
```text
haut : [01][02][03][04][05][06][07][08][09][10][11][12][13][14][15][16][17][18][19][20][21][22]
 [23][24]

 | |
 A15 /IRQ

bas : [48][47][46][45][44][43][42][41][40][39][38][37][36][35][34][33][32][31][30][29][28][27]
 [26][25]

 | | | | | | | | | | |
 | | | | | D0 D1 D2 D3 D4 D5
 5V GND 02 01 00

D6 D7
```
```

Recommended proto 1 safe bundle

This is now the ****recommended V2 proto 1 safe bundle****.

The idea is:

- keep the active protocol simple
- use `OUT0..OUT2` as command
- use `/OE2` plus the `PORT1-0..4` group as the read side for `\$4017`
- avoid starting with the full `CPU D0..D7` sniff bundle

```
```text
haut : [01][02][03][04][05][06][07][08][09][10][11][12][13][14][15][16][17][18][19][20][21][22]
 [23][24]

 | | | | | | | |
 GND A15 OE2 /IRQ P1-2 P1-3 P1-4

bas : [48][47][46][45][44][43][42][41][40][39][38][37][36][35][34][33][32][31][30][29][28][27]
 [26][25]

 | | | | |
 5V GND 02 01 00
```
```

Additional lines on the same safe proto bundle:

```
```text
`19 = PORT1-0`
`20 = PORT1-1`
```
```

So the ****safe first-choice wiring set**** is:

```
1. `48` = `+5V`
2. `47` = `GND`
3. `02` = `GND` secondaire
4. `45` = `OUT2`
5. `44` = `OUT1`
6. `43` = `OUT0`
7. `11` = `/OE2` (`$4017` read strobe)
8. `19` = `PORT1-0`
9. `20` = `PORT1-1`
10. `15` = `PORT1-2`
11. `16` = `PORT1-3`
12. `18` = `PORT1-4`
13. `14` = `/IRQ` optionnel
14. `05` = `A15` bonus debug
```

With a `16-wire ribbon`, the last `2` conductors stay available for:

```
15. spare / reserve
16. spare / reserve
```

Advanced bundle (more powerful, more risky)

This is the other direction we discussed, but ****not**** the recommended bring-up path:

```
- `CPU D0..D7`
- `A15`
- `/IRQ`
- later `R/W` and `M2`
```

Keep this for a later bus-aware proto, not for proto 1.

Signal list

Top row

```
- `02` = `GND`
- `05` = `A15`
- `11` = `/OE2` (`$4017` read strobe)
- `14` = `/IRQ` with `1 kOhm` series resistor
- `15` = `PORT1-2`
- `16` = `PORT1-3`
- `18` = `PORT1-4`
- `19` = `PORT1-0`
- `20` = `PORT1-1`
```

Bottom row

```
- `48` = `+5V`
- `47` = `GND`
- `45` = `OUT2`
- `44` = `OUT1`
- `43` = `OUT0`
- `32` = `CPU D0`
- `31` = `CPU D1`
- `30` = `CPU D2`
- `29` = `CPU D3`
```

- `28` = `CPU D4`
- `27` = `CPU D5`
- `26` = `CPU D6`
- `25` = `CPU D7`

Quick reality check before soldering all lines

- `02` and `47` should both be `GND`
- `43`, `44`, `45` should be the three contiguous `OUT0`, `OUT1`, `OUT2` lines
- in this **proto 1 safe** version, `11` is the important read strobe line for `\$4017`
- `19`, `20`, `15`, `16`, `18` are the five controller data lines we are actively using here

If that matches your meter, the orientation is locked.

Cable / connector note

For a first V2 bundle, we need roughly `16` lines.

What exists in standard parts:

- `16-conductor ribbon cable`
- `HE10 / IDC 16-pin connectors`
- `16-pin IDC board headers`

What probably does **not** exist as a neat off-the-shelf "perfect V2 cable":

- a ready-made round cable already terminated exactly for this project

So the realistic options are:

1. flat `16-way ribbon` with IDC/HE10 ends
2. ribbon cable that is separated for a short distance near the motherboard
3. custom round multicore cable only if we later want a cleaner final build

Sources

- https://www.nesdev.org/wiki/Expansion_port
- <https://commons.wikimedia.org/wiki/File:Nintendo-NES-Mk1-Motherboard-Bottom.jpg>
- <https://www.e44.com/connectique/connecteurs/fiches/IDC-16b/Femelle/he10-femelle-sertir-16-pin-s-HE10F16.html>
- <https://www.e44.com/connectique/connecteurs/fiches/DIN-3-broches/Male/he10-male-sertir-16-pin-s-pas-2.54mm-HE10M16.html>
- <https://www.e44.com/connectique/connecteurs/embases-ci/IDC-16b/Male/he10-male-droit-16-pins-H-E10MD16.html>
- <https://www.e44.com/cablages/cables/cables-en-nappe/cable-en-nappe-16-conducteurs-0.08mm2-pas-1.27mm-1m-gris-FNG16-1.html>
- <https://www.e44.com/cablages/cables/cables-en-nappe/>